

SHANMUKH KOMMU

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Associate Software engineer specializing in LLM-based automation, workflow agents, and enterprise RAG systems, with strong collaboration and communication skills in cross-functional environments.

EDUCATION

B.Tech. Artificial Intelligence and Data Science — KL University Hyderabad

Graduated: 2026 | CGPA: 9.42

SKILLS

Programming	Python(AsyncIO, Asynchronous Programming), SQL
Frameworks	PyTorch, TensorFlow, Django, Flask, Scikit-learn, FastAPI
ML/AI	LLMs, Langchain, RAG, Transformers (BERT, HuggingFace), MCP, LangGraph, AutoGen, Prompt Engineering, DeepEval, RAGAS
Tools	Git, Docker, Jenkins, CI/CD, Linux, PostgreSQL, MySQL, Kafka, Kubernetes, Keycloak, Celery, Redis
Data Formats, APIs	JSON, REST APIs, Websockets

EXPERIENCE

Associate Software Engineer

Techsophy

09/2025 - Present

Hyderabad, Telangana

- Engineered a **multi-camera person re-identification system** that reduced false positive matches by 73% in crowded environments. Leveraged 512-dimensional facial embeddings (**InsightFace**) and motion-based tracking (**DeepSort**) to enable security teams to track individuals across **8+ CP Plus** camera feeds with 94% accuracy
- Automated manufacturing date and batch number extraction** from product labels in warehouses, and reducing QC processing time by 18 hours/week through an OCR to transformer-based LLM pipeline that achieved 96% accuracy on degraded/rotated text, eliminating manual data entry for 300+ daily inspections. (**Paddle OCR, GPT-4o-nano.**)
- Built a real-time health monitoring pipeline using **Apache Kafka** to ingest and process high-volume patient vitals, structuring metrics into JSON schemas to construct a **digital twin**. Developed a **RAG agent** that indexes and chunks research journals as vectors in MongoDB and queries the digital twin to generate automated diagnosis. Implemented Redis for cache storage and integrated RAGAS for retrieval evaluation and Presidio for PII redaction on sensitive patient data.

PROJECTS

- PulseAI** [\[GitHub\]](#)
 - Built PulseAI, a Kafka-powered personal activity intelligence system that tracks real-time window activity, streams events through Apache Kafka, persists data to SQLite, and generates AI-powered productivity insights using a local LLM (Mistral via Ollama). Developed a Tkinter desktop GUI for one-click start/stop and report generation. Implemented a Jenkins CI/CD pipeline with automated pytest testing on Docker.
- Hallucination Insurance** [\[GitHub\]](#)
 - Designed and deployed a **local-first Hallucination Insurance API** that detects LLM hallucinations using a hybrid semantic similarity and LLM verification approach. Automatically extracts and verifies individual claims from LLM responses against source documents using Sentence-Transformers and ChromaDB vector search, with Mistral (via Ollama) handling ambiguous cases. Built with FastAPI backend and Streamlit UI.

CERTIFICATIONS

AWS Certified AI Practitioner – Amazon Web Services

[Credential](#)

GitHub Foundations Certification – GitHub

[Credential](#)